Welcome to .NET 9



Build anything with a unified platform

.NET

Cloud | Web | Desktop | Mobile | Gaming | IoT | AI

Tools


Visual Studio


Visual Studio Code


CLI


GitHub Copilot

+

Operating system


Windows


Linux



macOS

+

Ecosystem


NuGet


GitHub


.NET Aspire


Components, tools, library vendors

.NET Aspire
Official Launch

Monthly
updates

OpenAI
OpenAI Library for .NET

Commitment to Security

Unity for VS Code

Improved GitHub
Copilot for .NET

C# Dev Kit

.NET MAUI
for VS Code

Ubuntu
Ubuntu 24.04

Since Last Year

160
Virtual Events
& Live Streams

45%
NuGet
Download
Increase

Smart
Components

.NET Community
Toolkit + Native AOT

.NET Aspire
Training

NuGet Audit 2.0

Native Library
Interop

Azure AI
Inference SDK

39%
.NET 8
Downloads
Over .NET 6

NuGet

Fastest Growing Package Ecosystem

420K+

Packages

570+ Billion

Downloads

45%

Download Increase

Security Features



OpenSSF
Pinned Dependencies
Trusted Publishers

GitHub
Dependabot 2.0

Deps.dev
Support

Loved by developers
Trusted by enterprises

6.5+ Million
Monthly active users

60,000+
Community members
have contributed to .NET

#1 2024 stackoverflow
Most Admired Developer Framework

Top 5 CLOUD NATIVE COMPUTING FOUNDATION
Highest velocity OSS project

Top 5
Top Programming Language on GitHub

https://survey.stackoverflow.co/2024/technology/#1-programming-scripting-and-markup-languages
https://survey.stackoverflow.co/2024/technology/#1-web-frameworks-and-technologies
https://survey.stackoverflow.co/2024/technology/#1-other-frameworks-and-libraries

https://github.blog/news-insights/research/the-state-of-open-source-and-ai/#the-most-popular-programming-languages

CNCF (Commits+PRs+Issues)
https://github.com/cncf/velocity/blob/main/reports/top_projects_by_activity.txt

.NET powers Microsoft at scale



aka.ms/dotnet-developer-stories

Microsoft 365

From .NET 6 to .NET 8*

- **35%** CPU usage
- **30%** memory consumption



Copilot

Re-written in C# with .NET 8

- **90%** smaller codebase
- **3X** performance
- **99.999%** reliability on day 1

XBOX

From .NET Framework to .NET 8**

- **15%** CPU time
- **13%** aggregated core time per query

* Core services that power Microsoft 365
** Services that power Xbox Live


ups


GE Aviation


CHIPOTLE
MEXICAN GRILL




H&R
BLOCK


SPORTS
NEXT
SPORTS x TECH


stackoverflow


Tencent 腾讯


SIEMENS
Healthineers


Cincinnati
Children's
changing the outcome together


GEOCACHING


Irth


Vestas


STADIO


Fidelity
INVESTMENTS


Allegiance
CONSULTING


HMRI
Hunter
Medical
Research
Institute


WILLIAMS
MULLEN
findingyes


DELHAIZE SERBIA


nopCommerce


STOPWATCH


SETPOINT
MEDICAL


AVEVA


LightSail
The Literacy Solution


IGNITION
GROUP


BurnRate


Red-Point


SPAREPARTS
NOW


endjin


escola ágil

Announcing .NET 9

aka.ms/get-dotnet-9



Productive



Modern & Secure



Intelligent



Performance

.NET 9 Highlights



Productive

.NET Aspire for improved developer inner loop

Deeper GitHub Copilot integration in Visual Studio

Hot Reload enhancements in VS Code with C# Dev Kit

Refreshed build output highlighting



Modern & Secure

Simplified Blazor authentication

Improved integration with cloud-native services & OpenTelemetry

Enhanced UI controls across Blazor, .NET MAUI, WinUI 3, WPF, and Windows Forms

dotnet restore auditing for known security vulnerabilities



Intelligent

.NET AI Building Blocks with Microsoft.Extensions.AI

AI Tokenizers

Tensor<T>

TensorPrimitives

.NET SDKs for vector databases

Server-Sent Event Parser



Performance

Adaptive Garbage Collection

Up to 25% faster Blazor startup

Optimized static web assets

with ref struct improvements

More Native AOT

Trimming optimizations

New NuGet restore resolver for large projects

Most performant release yet

>1000 Performance-related PRs

JSON

Native AOT

JIT

Compression

PGO

ARM64

Bounds Check

ARM SVE

Cryptography

Vectorization

Write Barriers

GC

Threading

Reflection

Object Stack Allocation

Loops

LINQ

Diagnostics

Numerics

Span

Inlining

Networking

https://devblogs.microsoft.com/dotnet/performance-improvements-in-net-9/

.NET Aspire

AI + .NET

.NET MAUI

ASP.NET Core

Developer Productivity

Biggest release of .NET yet

.NET 9 Community

>26k
Contributions

>9k
Community members



Thank You!

Best way to build apps

Modern Workloads


Web


Client


Artificial Intelligence


Cloud

Developer Productivity


Language


Libraries


Developer Tools


Modernization

Fundamentals


Observable


Performance


Secure by Design


Resilient


Accessible


Scalable


Manageable

.NET Aspire

Your building blocks for app development





.NET Aspire

Build, test, and deploy apps seamlessly from code to cloud

- Streamlined Inner-Loop
- Developer Dashboard
- Integrations
- Deployment

Extensible, OpenTelemetry Built-in, & Ready for Any Cloud



Get started

Open-source

Templates

Integrations

To build

Service discovery

Developer dashboard

Logs, metrics, distributed traces

To deploy

Single command run

App topology in C#

Cloud deployment

Demo







Announcing

.NET Aspire Community Toolkit

github.com/CommunityToolkit/aspire







Preview

Azure Functions support for .NET Aspire



".NET Aspire has really helped us tighten our inner development loop"



Logs



Metrics



Traces



Azure Integrations

Craig Taylor
Principal Architect, XBOX Live

.NET Aspire 9.0

Azure Functions preview support

More secure defaults

"WaitFor" resources to spin up

Start/Stop Resources

Improved Azure Configuration

Simplified acquisition

Improved Azure Container Apps integration

Persistent Containers

Support for .NET 8 & .NET 9

OpenAI Integration

AWS Stable Integrations

Visual Studio & C# Dev Kit enhancements

.NET Aspire Community Toolkit

Tune in at **9:00am** PST for What's New in .NET Aspire

AI

Integrating AI into your .NET app

.NET & AI



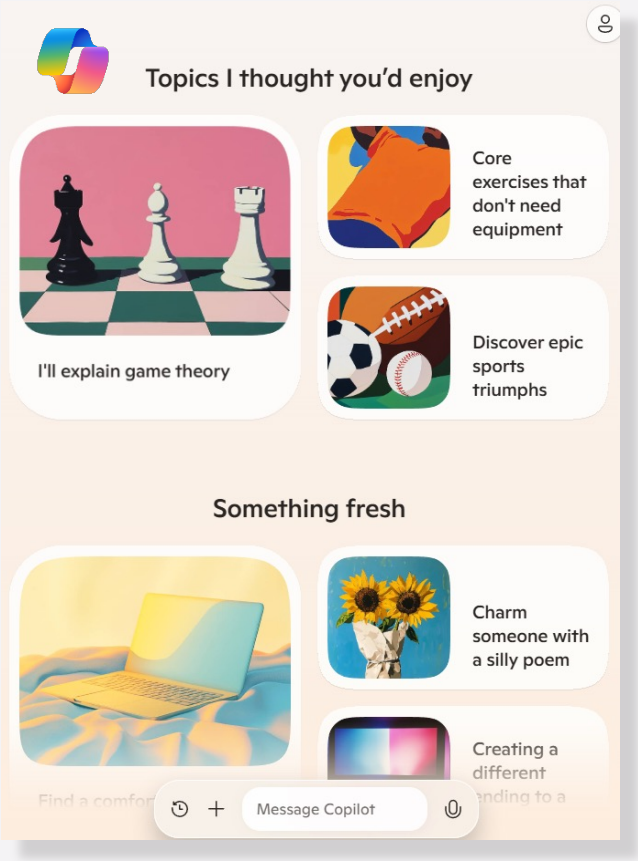
Learn

Integrate

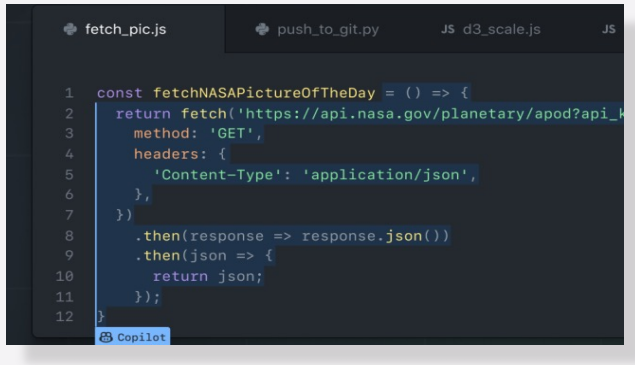
Collaborate

Deploy

Generative Applications Built on .NET



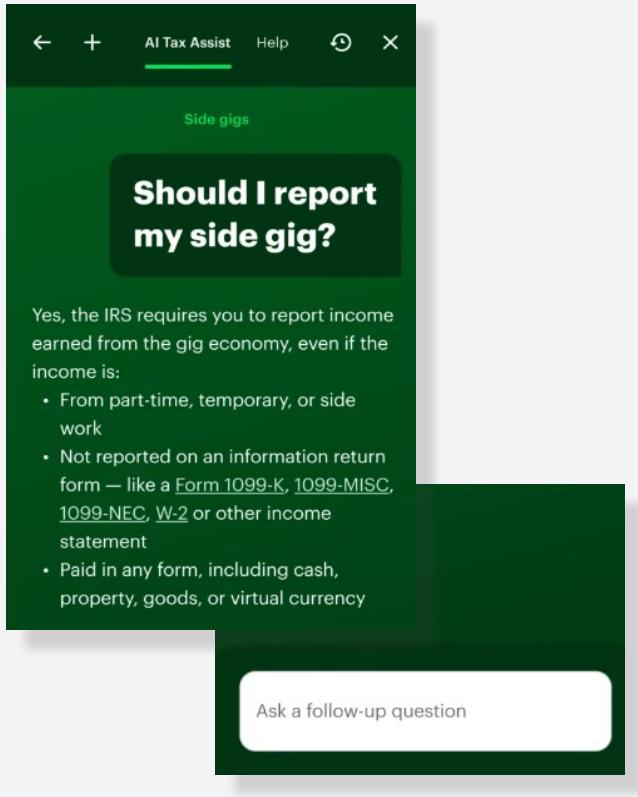
Microsoft Copilot



Copilot assisted coding

GitHub

GitHub Copilot



H&R Block



KPMG Clara

"Our .NET stack supported rapid development of these AI features with the language and tools our developers know and love"

- Thomas Mackenzie, Global Audit Chief Technology Officer

KPMG Clara with AI empowers our **95,000 audit professionals to improve audit quality and focus on higher risk areas**.

Built with .NET, AI, & Azure

.NET AI Ecosystem



Semantic Kernel



Azure OpenAI



Azure AI Search



Azure Inference



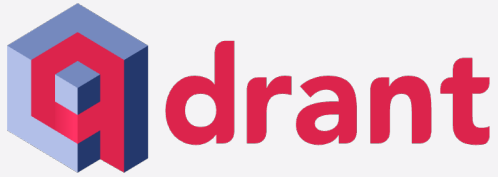
OpenAI library for .NET



LlamaParse C# Client



GitHub Models



Qdrant Solutions



C# Client



C# Client



Smart Components



Smart Components



Smart Components

Microsoft.Extensions.AI

.NET AI Library

Streamline AI integration with our unified APIs

Common AI Abstractions

Standard Middleware

Interoperability and Extensibility

Available in Preview Today



PIECES®

FOR DEVELOPERS

Pieces: an AI productivity toolchain for developers.

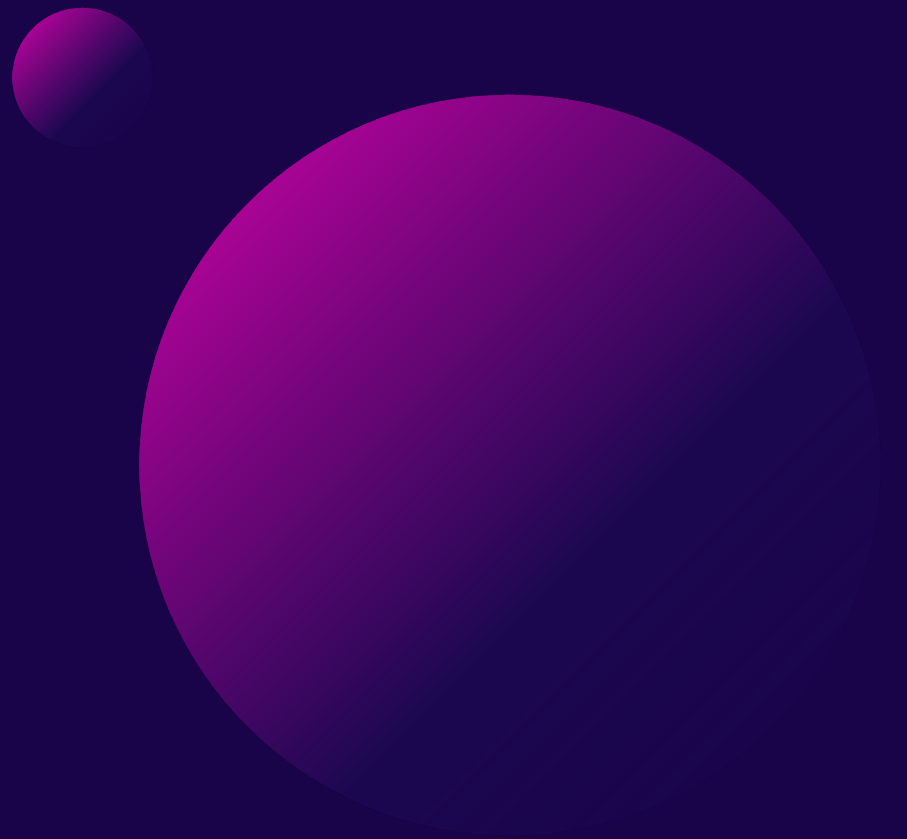
“Our team was able to integrate Microsoft.Extensions.AI into our SDK in just a few hours, providing our SDK users a simpler way to integrate the **long-term memory** of Pieces into their .NET Applications”

Using Microsoft.Extensions.AI

Demo

Microsoft.Extensions.AI

Available in Preview



Azure AI Inference SDK for .NET

TensorPrimitives

Smart Components

AI App Gallery

Microsoft.Extensions.AI Library Available in **Preview Today**

.NET and AI

Pinecone Official .NET Client

.NET Aspire Integrations for AI

Microsoft Learn Training

New .NET Samples

AI Tokenizers

OpenAI

Official OpenAI library for .NET

Tune in at **9:45am** PST for AI Building Blocks – A new, unified AI layer

Client App Development



Microsoft's investments in client development with .NET

Windows Native





WinUI3



WinForms



WPF

Cross Platform Native





.NET MAUI

Hybrid



Blazor Hybrid



WebView2

Web





Blazor Web

.NET Multi-platform App UI (MAUI)

Single project system, single codebase

WinUI



Mac Catalyst



iOS



Android



.NET MAUI in .NET 9



Controls



App Lifecycle



Performance



Platform Integrations

.NET MAUI is Active and Growing



>30%

Active users

1600+ PRs,
600+ Issues Closed
in .NET 8

1900+ PRs,
2800+ Issues Closed
since .NET 8 release

github.com/dotnet/maui



Syncfusion Toolkit for .NET MAUI

14 free, open-source controls available for .NET MAUI!

Open-source on github.com/syncfusion/maui-toolkit

Package Syncfusion.Maui.Toolkit available on NuGet

Syncfusion

Demo

Retail Insight

3M

hogeschool
vives

Cincinnati
Children's

POLARIS

Xari

Irth

OEXA

MAHIZ
grow your business

HAWAIIAN
AIRLINES

ups

CARDIFF
CAERDYDD

Fidelity
INVESTMENTS

Microsoft
Dynamics 365

H3Y

Microsoft
Azure

CIVICA

SPORTS
NEXT
SPORTS x TECH

FINLOCKER

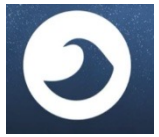
Belgian
Red Cross
Flanders

ambientindustries

A

Medtronic

insight

esri


Soluzione 1

DevInMotion

REY
TECHNOLOGY

[A]


escola ágil

.NET MAUI in .NET 9

Android 35 & iOS 18 Support

Xcode Sync CLI

iOS Multi-target bindings

Native Library Interop

New lifecycle events

Activate Window

CollectionView & CarouselView Performance Improvements

New TitleBar Control

android Asset Packs

Native Embedding Enhancements

New Input Keyboards

.NET Native AOT improvements

Trimming Enhancements

HybridWeb View

Justify text alignment

Tune in at **12:00pm** PST for What's New in .NET MAUI

Web Development



ASP.NET Core:

Your Full Stack Web Platform for .NET



Web UI



APIs



Data



Server



Security



Tooling

ASP.NET Core in .NET 9



Quality & fundamentals



Developer experience



Cloud native

The fastest ASP.NET Core yet!

- ⬆ **15%** throughput*
- ⬇ **93%** memory usage on high core-count machines*
- ⬇ **25%** startup time for Blazor WebAssembly**

WebSocket message compression for Blazor Server

Precompression and improved caching for static web assets

* TechEmpower JSON Min APIs, Intel Gold 56 cores (logical) Linux
** Lighthouse

Demo



ASP.NET Core in .NET 9

Optimized static web asset handling

Improvements to exception handling and debugging

Authentication enhancements

New Blazor Hybrid Templates

Improved Kestrel connection metrics

SignalR improved distributed tracing

Dictionary debugging improvements

Built-in OpenAPI support

Detect Blazor component render mode

Improvements to DataProtection

SignalR AOT support

Blazor reconnection improvements

Trust Developer certs on Linux

Up to 25% faster Blazor startup

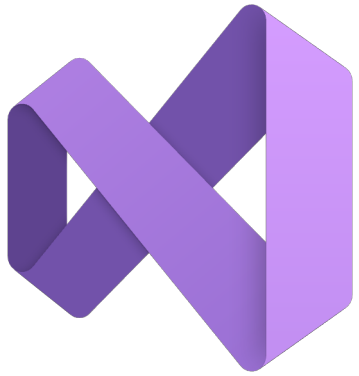
Keyed service support in middleware

What's New for ASP.NET Core & Blazor in .NET 9

API-ly Ever After: OpenAPI in .NET 9

Introducing HybridCache in ASP.NET Core

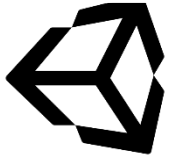
Developer Productivity



Visual Studio



Visual Studio Code









GitHub Copilot

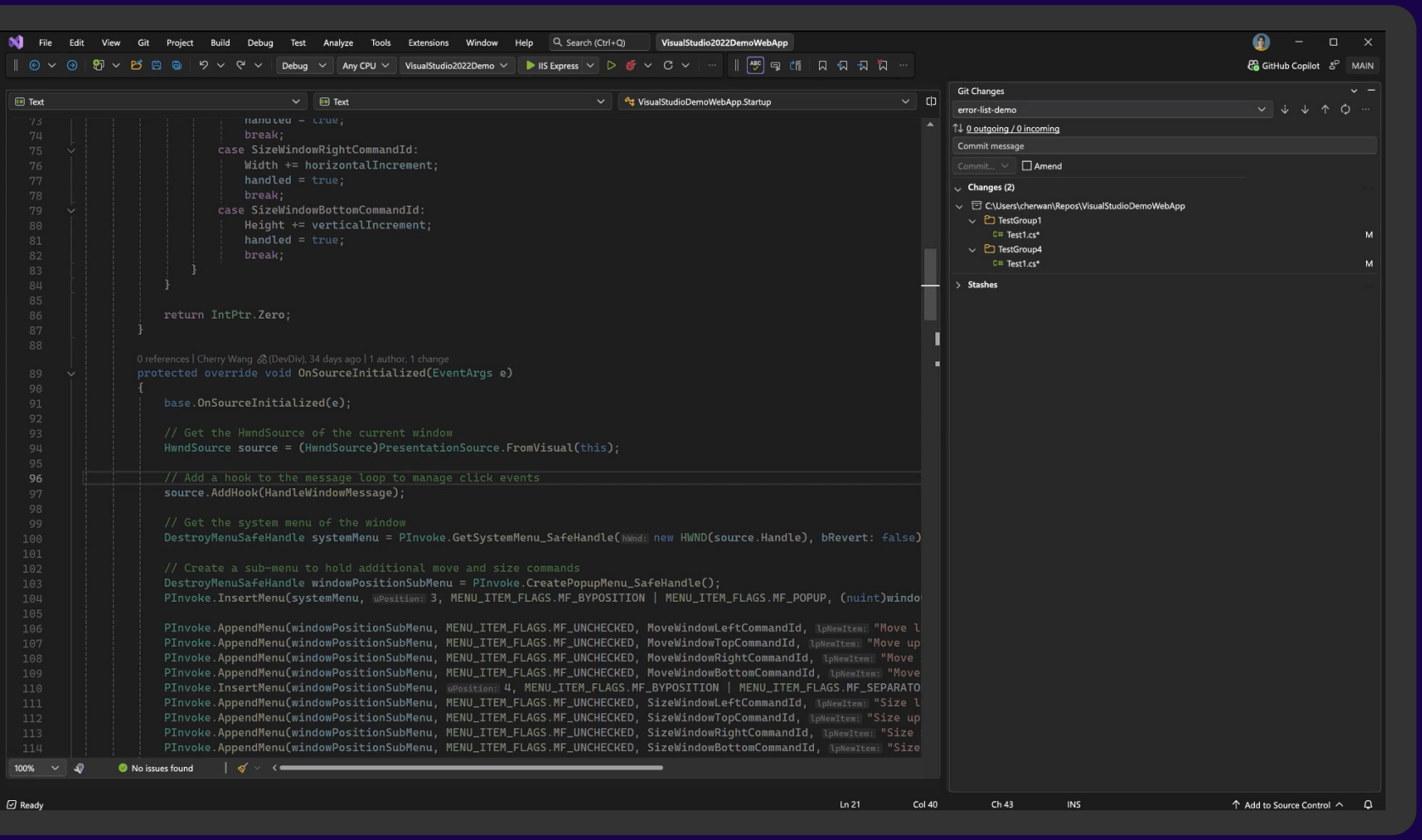
Demo

GitHub Copilot



Visual Studio 2022 v17.12

Released



Upgraded GitHub Copilot experience

Modern development for .NET 9 and .NET Aspire

More user-requested features than ever before

Available today: **visualstudio.com/download**

.NET Aspire

AI + .NET

.NET MAUI

ASP.NET Core

Developer Productivity

ASP.NET Core Security Diagnostics	Blazor Authentication Enhancements	Tensor Primitives	AI Tokenizers	Binary Formatter removal	Microsoft. Extensions.AI	C# 13	Enhanced Metrics	HybridCache
Azure Container App Production Diagnostics	Virtual Exceptions	Optimized Static Web Asset Handling	Adaptive Garbage Collection	Persistent Containers	Simplified Acquisition	Workload History	"WaitFor" Resources to Spin Up	
Refreshed Build Output Highlighting	dotnet restore Auditing	New & Refreshed UI Controls	Microsoft Learn Training	Android 35 & iOS 18 Support	Trimming Enhancements	.NET Aspire Community Toolkit	F# 9	ARM Performance
with ref struct Improvements	Up to 30% Faster Blazor Startup	EF Core Support for CosmosDB	Server-sent Events Parser	SDKs for Vector Databases	Visual Studio & C# Dev Kit Enhancements	Enhanced Native AOT for EF Core	CollectionView & CarouselView Performance Improvements	

Your Windows Apps on .NET 9

Performance	Accessibility	Modern OS Features	Latest Language Features	Improved Tooling	Developer Productivity



WinUI 3

Native AOT support
WebView2 via NuGet
Package Deployment APIs

microsoft / Microsoft-ui-xaml
microsoft / WindowsAppSDK

.NET WPF

Fluent UI Theming
Theme Mode
Accent Colors
Hyphen based ligatures

dotnet / wpf

.NET WinForms

APIs for modern icons
Control.InvokeAsync
Dark Mode

dotnet / winforms

Get Modern & Stay Modern

Chevron Phillips CHEMICAL

Performance by design.
Caring by choice.™

"We can now rapidly develop prototypes in Python and .NET and securely deploy them at scale"– Melba Swapna, Cloud Architect Manager

.NET Upgrade Assistant
aka.ms/upgrade-assistant

Azure Migrate: application & code assessment
aka.ms/appcat



Scale with Azure

Ready for .NET 9

Check service announcements for full availability.

Get .NET 9

aka.ms/get-dotnet-9

visualstudio.com/download

Learn and get certified

freeCodeCamp(🔥)

aka.ms/csharp-certification

Microsoft Learn

aka.ms/aspire-credential

.NET 9 and Visual Studio 2022 updates available today

Thank you!

